ResearchOS Multi-Agent Research Architecture

Human-guided, evidence-grounded research from source ingestion to verified academic output

Researcher and Project Workspace

research goal, files, constraints

External Services
Model providers
Scholarly search and tools

LLM · API

models and tools

RESEARCHOS RUNTIME

1 · CONTEXT AND ORCHESTRATION

Context Compiler
Draft, prompt, sources, and history

Dynamic Planner
Decompose, prioritize, and schedule

Agent Coordinator
Route, monitor, retry, and replan

Policy and Guardrails
Scope, budgets, and approvals

2 · SPECIALIZED RESEARCH AGENTS

Literature Agent
- Discover and deduplicate studies
- Build evidence-backed syntheses
- Map the research landscape

Gap and Hypothesis
- Identify unresolved questions
- Generate falsifiable hypotheses
- Track hypothesis lineage

Novelty and Feasibility
- Compare prior art and contribution
- Assess data, compute, and risk
- Rank viable research directions

Experiment and Statistics
- Design and execute analyses
- Review methods and uncertainty
- Capture reproducible provenance

Critic and Writing
- Challenge claims and assumptions
- Revise evidence-linked content
- Compose academic sections

3 · SHARED EVIDENCE, VERIFICATION, AND DELIVERY

Hybrid Retrieval
- Dense and lexical search with reranking
- Page-level provenance and evidence packs

Shared Research State
- Claims, sources, citations, and artifacts
- Agent outputs with complete provenance

Verification Layer
- Claim-to-evidence and citation checks
- Contradictions, quality, and completeness

Research Outputs
- Draft, chat, and cited references
- Word and PDF export

PERSISTENT PLATFORM

PostgreSQL projects and runs

Vector Index chunks and embeddings

Artifact Store sources, results, and exports

Security and Audit tenant scope, approvals, and usage

Navy: workflow · Teal: evidence flow · Coral dashed: quality-control feedback · Gray dashed: coordination and persistence